

1. Concept of Adjustable Threshold  $\theta$ :

a simple perceptron:

$$\Rightarrow z = \sum_0^n w_i \cdot x_i = w_0 \cdot x_0 + w_1 \cdot x_1 + \dots + w_n \cdot x_n$$

$$y = f(z), f(z) = \begin{cases} -1, & z < 0 \\ +1, & z \geq 0 \end{cases}$$

we can make the perceptron more general by replace activation function as follow:

$$\Rightarrow z = \sum_0^n w_i \cdot x_i = w_0 \cdot x_0 + w_1 \cdot x_1 + \dots + w_n \cdot x_n$$

$$y = f(z), f(z) = \begin{cases} -1, & w_1 \cdot x_1 + \dots + w_n \cdot x_n < -w_0 \cdot x_0 \\ +1, & w_1 \cdot x_1 + \dots + w_n \cdot x_n \geq -w_0 \cdot x_0 \end{cases}$$

$\Rightarrow$  set bias weight ( $w_0$ ) =  $-\theta$ , and bias input ( $x_0$ ) is always 1

$$\Rightarrow z = \sum_0^n w_i \cdot x_i = w_0 \cdot x_0 + w_1 \cdot x_1 + \dots + w_n \cdot x_n$$

$$y = f(z), f(z) = \begin{cases} -1, & w_1 \cdot x_1 + \dots + w_n \cdot x_n < \theta \\ +1, & w_1 \cdot x_1 + \dots + w_n \cdot x_n \geq \theta \end{cases}$$

$$\Rightarrow z = \sum_0^n w_i \cdot x_i = w_0 \cdot x_0 + w_1 \cdot x_1 + \dots + w_n \cdot x_n$$

$$y = f(z), f(z) = \begin{cases} -1, & -\theta + w_1 \cdot x_1 + \dots + w_n \cdot x_n < 0 \\ +1, & -\theta + w_1 \cdot x_1 + \dots + w_n \cdot x_n \geq 0 \end{cases}$$

$$\Rightarrow z = \sum_0^n w_i \cdot x_i = w_0 \cdot x_0 + w_1 \cdot x_1 + \dots + w_n \cdot x_n$$

$$y = f(z), f(z) = \begin{cases} -1, & w_0 \cdot x_0 + w_1 \cdot x_1 + \dots + w_n \cdot x_n < 0 \\ +1, & w_0 \cdot x_0 + w_1 \cdot x_1 + \dots + w_n \cdot x_n \geq 0 \end{cases}$$

$$\Rightarrow z = \sum_0^n w_i \cdot x_i = w_0 \cdot x_0 + w_1 \cdot x_1 + \dots + w_n \cdot x_n$$

$$y = f(z'), f(z') = \begin{cases} -1, & z' < 0 \\ +1, & z' \geq 0 \end{cases}$$

now, using the new activation function  $f(z')$  create an adjustable threshold perceptron

$$\because z' = w_0 \cdot x_0 + w_1 \cdot x_1 + \dots + w_n \cdot x_n = -\theta + w_1 \cdot x_1 + \dots + w_n \cdot x_n = +w_1 \cdot x_1 + \dots + w_n \cdot x_n - \theta$$

$$\because \text{if } w_1 \cdot x_1 + \dots + w_n \cdot x_n < \theta \longrightarrow z' < 0 \longrightarrow f(z') = -1$$

$$w_1 \cdot x_1 + \dots + w_n \cdot x_n \geq \theta \longrightarrow z' \geq 0 \longrightarrow f(z') = +1$$

2. Role of Bias Term  $w_0 \cdot x_0$  of Perceptron:

the bias term  $w_0 \cdot x_0$  of perceptron is simply the same as  $b$  ( $y$ -intercept) in the equation for a straight line " $y = mx + b$ ".